

A novel active equalization method for lithium-ion batteries in electric vehicles



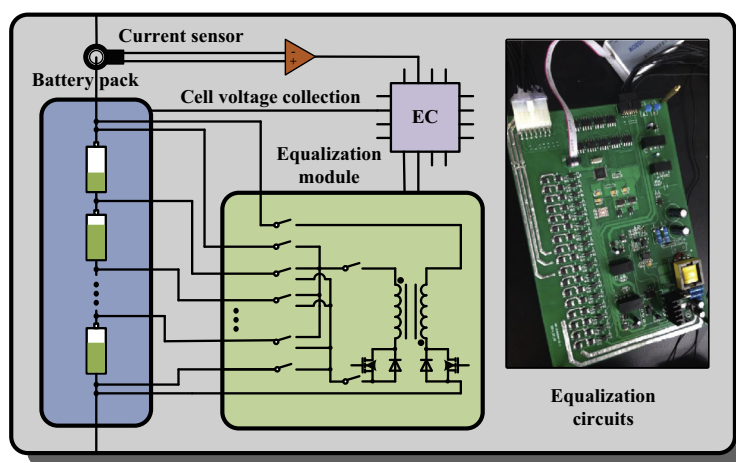
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HIGHLIGHTS

- Build an active equalization method for lithium-ion batteries.
- A bidirectional transformer topology is introduced for active equalization.
- The PF method is used for cell SOC estimation to eliminate drift noise of current.
- The SOC based equalization algorithm is analyzed with different SOC bounds.

GRAPHICAL ABSTRACT



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ABSTRACT

Cell inconsistency is inevitable due to manufacturing constraint. Therefore, cell equalization is essentially required. In this paper, we propose a novel active equalization method based on the remaining capacity of cells which is feasible for lithium-ion battery packs in electric vehicles (EVs). The cell models are established based on a combined electrochemical model of lithium-ion batteries. The remaining capacity and state-of-charge (SOC) of cells are observed at the beginning of equalization. The particle filter (PF) method is employed to estimate the cell SOC during equalization in order to eliminate the drift noise of the current sensor. The first high-SOC cell discharge (FHCD) and first low-SOC cell charge (FLCC) equalization algorithms are proposed and compared with 1% and 3% SOC bounds, respectively. The validation experiment results have shown that the proposed algorithm is suitable for equalization of lithium-ion batteries in EVs.

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1. Introduction

Anxieties of environment deterioration and global energy resources depletion force traditional vehicles to be cleaner. Electric

vehicles (EVs) are gradually accepted by people as they are zero-emission and environmentally friendly transports. In EVs, hundreds to thousands of cells are connected in series and parallel to provide sufficient power. Inconsistencies in cells are concerned due to the inconsistent manufacturing process and the inhomogeneous operating environment [1]. Significant degradation of maximum release capacity, energy density and life cycles can be observed

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after pack construction due to cell variations [2]. Therefore equalization technology is necessary for the energy or battery management system (EMS/BMS).

In recent years, the equalization topologies and algorithms for lithium-ion battery packs are studied in order to mainly keep the remaining capacity of the cells balanced and extend their lifetime [3–8]. The equalization topologies can be divided into two main groups: passive equalization methods and active equalization methods. The fixed shunting resistor method is one of the most common used passive method which uses a resistor in parallel with each individual cell. The current is partially bypassed from the cells in order to limit the cell voltage. Therefore current is continuously bypassed and battery energy is continuously wasted through this method. The active equalization methods are proposed which use external circuits to actively transport the energy among cells [3,4].

Regardless of the equalization topologies, appropriate equalization algorithms are required to maximize the equalization efficiency. According to the literatures [5–9], the equalization algorithms can be divided into two categories: voltage based and state-of-charge (SOC) or capacity based equalization algorithms. Voltage based equalization algorithms [5–7] are extensively adopted in most real-time systems. The SOC based equalization algorithms [8,9] require accurate remaining capacity and cell SOC estimation which is more suitable for lithium-ion batteries. This is because the charge/discharge curves of lithium-ion batteries are nonlinear, a small voltage variation may have a large capacity inconsistency especially at the flat charge/discharge plateau. Many methods have been proposed in literatures for accurate SOC estimation. Plett [10–12] applied the EKF method for SOC estimation and believed that high-fidelity cell models are required to estimate accurate SOC. He et al. [13] proposed a new working model that takes the drift current as a state variable to eliminate the effects of drift noise in SOC estimation. Zhong et al. [9] analyzed the relationship between the pack SOC and the in-pack cells under different equalization conditions and proposed a method for battery pack SOC estimation. Liu et al. [14] developed a temperature-compensated battery model and employed a dual particle filter for SOC estimation. More researches on accurate SOC estimation can be found in our previous works [15–20].

Along this paper, a novel active equalization method is proposed for the lithium-ion batteries in EVs based on the

SOC-Particle Filter (SOC-PF) algorithm. This paper is organized as follows: In Section 2, 12 series of LiFePO₄ cells are chosen from an urban electric bus. The cell inconsistencies are analyzed with a dynamic charging current profile. In Section 3, a bidirectional active equalization topology is first introduced. We subsequently propose a SOC based equalization algorithm based on particle filter. In Section 4, the first high-SOC cell discharge (FHCD) and first low-SOC cell charge (FLCC) equalization algorithms are proposed and compared with 1% and 3% SOC bounds, respectively. The numerical results have shown that the proposed algorithms are suitable for equalization of lithium-ion batteries in EVs.

2. Lithium-ion cell inconsistency analysis with dynamic charging current profile

In EVs, equalization circuits are used to discharge the cells with higher SOC to the battery pack and charge the cells with lower SOC from the whole battery pack to extend range and avoid early cut off. Identification of cell voltage and SOC inconsistency for lithium-ion battery pack is challenging due to the relatively flat SOC-OCV curves of lithium-ion cells compared to other kinds of batteries [1]. We experimentally investigate the cell voltages of 12 series IFP1865140-type cells from an urban electric bus which are developed by Hefei Guoxuan High-Tech Power Energy CO. Ltd. of China. The cells are selected to compose a small battery pack for experimental investigation with the nominal capacity of 12.5 A h. The dynamic charging current profile is shown in Fig. 1(a). Fig. 1(b) shows the pack SOC which is determined by the maximum remaining capacity cell during charging and minimum remaining capacity cell during discharging.

As can be seen from Fig. 1(c) and (d) the cell charging voltages (CCVs) of 12 series cells are different during the charging process. The average of maximum voltage difference during charging and resting are 15 mV and 5 mV. It is worth noting that the total voltage variation from 20% to 80% SOC is usually under 0.2 V. Meanwhile, the dynamic voltage varies dramatically when the charging current changes. The CCVs based equalization method is easy to cause over-equalization due to the flat OCV curve and voltage plateaus in large SOC ranges. Therefore we propose a SOC based equalization algorithm which is more applicable to lithium-ion batteries.

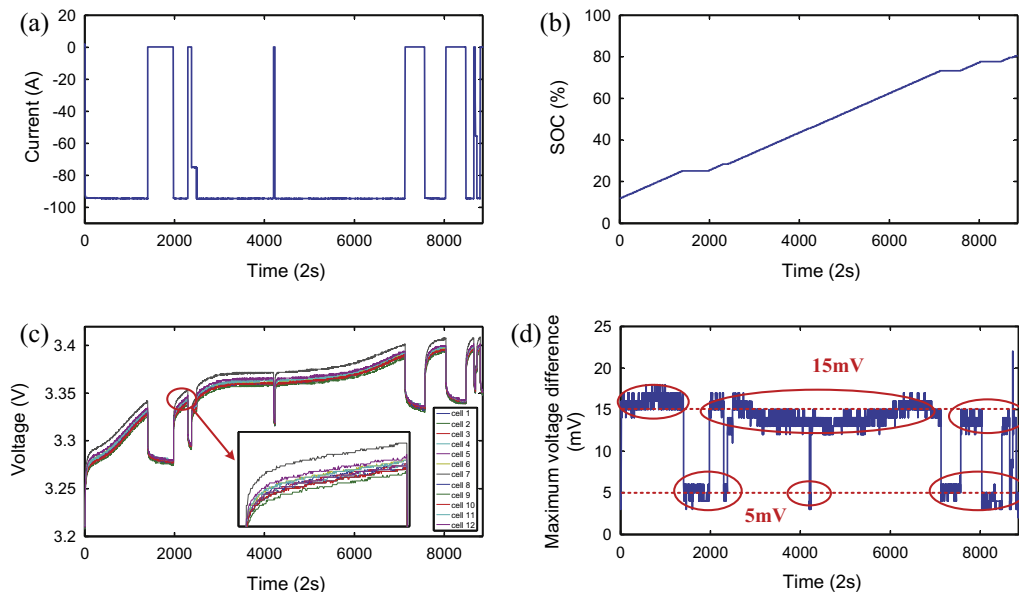


Fig. 1. Lithium-ion battery charging experiments: (a) dynamic charging current. (b) Pack state-of-charge. (c) CCVs of 12 series batteries. (d) Maximum voltage difference of the 12 series cells.

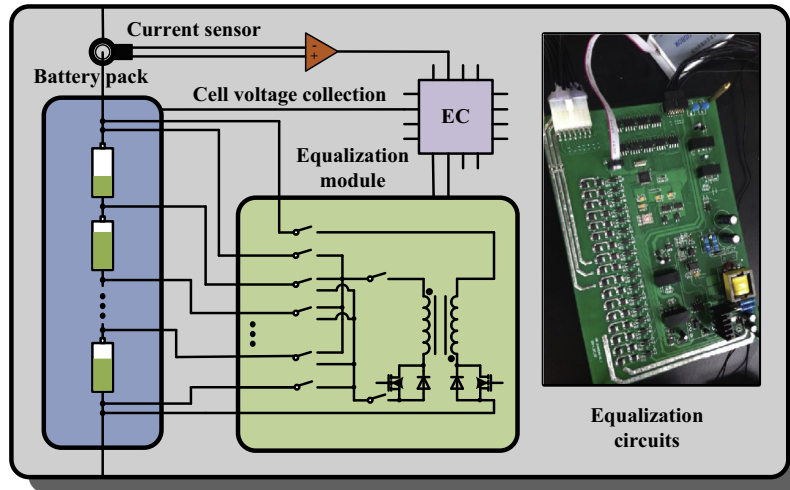


Fig. 2. Block diagram for equalization.

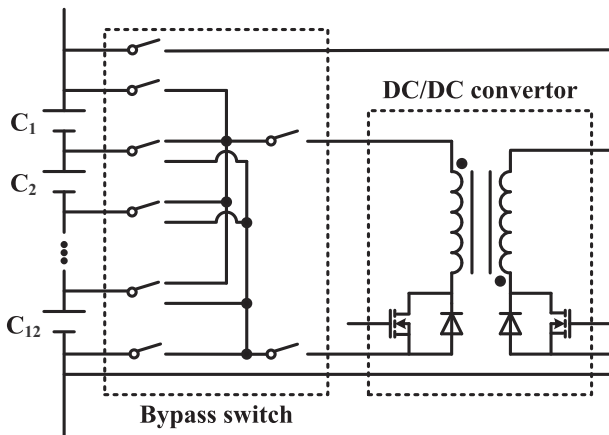


Fig. 3. Bidirectional active equalization topology.

3. Active equalization topology and SOC based equalization algorithm

3.1. Bidirectional active equalization topology

The block diagram of the active equalization system is shown in Fig. 2. The battery pack is composed by 12 IFP1865140-type cells. The loop current is measured by a FS500E2-type hall current sensor with a $\pm 1\%$ FS (full scale: ± 500 A) measurement error and the

equalization currents are measured by the hall effect-based linear current sensor ACS714 with a typical $\pm 0.075\%$ FS (full scale: ± 20 A) measurement error. The cell voltages are measured by the equalization controller (EC) with a 12 bit internal analog–digital converter and the sampling rate is 10 ms. A bidirectional active equalization circuit is designed as shown in Fig. 3 which allows the cell to pack and pack to cell equalization. The bypass switches and the bidirectional DC/DC converter are controlled by the EC. The cell to pack equalization, in case the cell to be equalized has a higher SOC than the reference one. The reference SOC is usually the threshold (0.5–1.5% higher than the average SOC). The pack to cell equalization, in case the cell to be equalized has a lower SOC than the threshold which is 0.5–1.5% lower than the average SOC.

3.2. Cell and battery pack models

Whether we choose the cell to pack equalization or the pack to cell equalization, accurate and reliable SOC estimations of cells are required. The SOC reflects the residual capacity of the cells, and are defined as the ratio of the remaining capacity to the total capacity. The cell SOC can be expressed by Eq. (1):

$$\text{SOC}(t) = \text{SOC}(t_0) + \eta_c \int_{t_0}^t (i(\tau) + i_E(\tau)) d\tau / C_N \quad (1)$$

where $\text{SOC}(t)$ is the SOC value at time t , $\text{SOC}(t_0)$ is the SOC value at initial time t_0 , C_N represents the nominal capacity, $i(\tau)$ represents the current at time τ , $i_E(\tau)$ represents the equalization current at

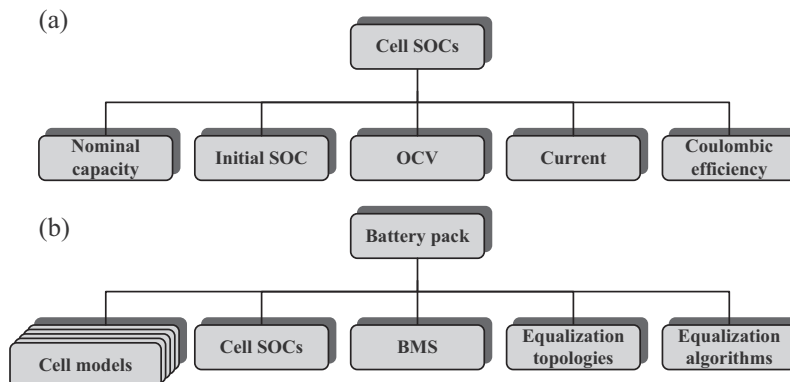


Fig. 4. Block diagrams for cell SOCs and battery pack: (a) cell SOCs. (b) Battery pack.

Table 1

Algorithm of the particle filter for SOC estimation.

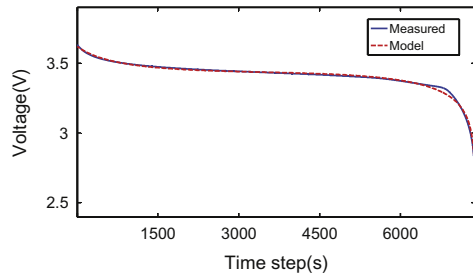
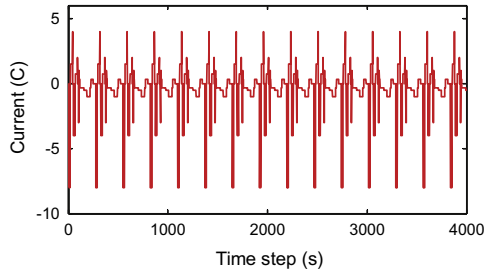
1. Initialization: $k = 0$
Randomly generate n initial particles x_0^i ($i = 1, 2, \dots, n$) for SOC
2. For $k = 1, 2, \dots$
State estimation for SOC:
 - (1) Importance sampling: The particle weight is calculated according to the follow equation:

$$w_k^i = \exp\left(-\frac{1}{2R}(y_k - y_k^i)^2\right) \cdot \frac{1}{\sqrt{2\pi R}}$$
 - (2) Normalize the importance weights as: $w_k^i = w_k^i / \sum_{i=1}^n w_k^i$
 - (3) Evaluate the effective sample size $N_{\text{eff}} = 1 / \sum_{i=1}^n (w_k^i)^2$ to judge the necessity of resampling
 - (4) The SOC of after completion of the algorithm is: $\text{SOC}_k = \sum_{i=1}^n w_k^i x_k^i$

Table 2

Parameters of the IFP1865140-type cell.

Parameter	Value
E_0	3.5354 V
K_0	0.5052 V
K_1	4.2293×10^{-5} V
K_2	0.1453 V
K_3	-0.1421 V
R	0.0247 Ω

**Fig. 5.** Comparison of measured and estimated voltage.**Fig. 6.** Dynamic current condition.

time τ , η_c represents the coulombic efficiency ($\eta_c \approx 1$ for LiFePO₄ batteries).

With the SOC available as part of the model state based on the electrochemical model, the open-circuit voltage can be expressed by Eq. (2) according to Ref. [11]:

$$\begin{aligned} \text{OCV}(t) &= E(t) + (i(t) + i_E(t))R \\ &= E_0 - K_0 \text{SOC}(t) - K_1 / \text{SOC}(t) + K_2 \ln(\text{SOC}(t)) \\ &\quad + K_3 \ln(1 - \text{SOC}(t)) \end{aligned} \quad (2)$$

where $\text{OCV}(t)$ represents the open-circuit voltage at time t , $E(t)$ represents the terminal voltage at time t , R represents the internal resistance, $i(t)$ represents the current at time t , E_0 represents the initial open-circuit voltage and K_i ($i = 0, 1, 2, 3$) are model coefficients.

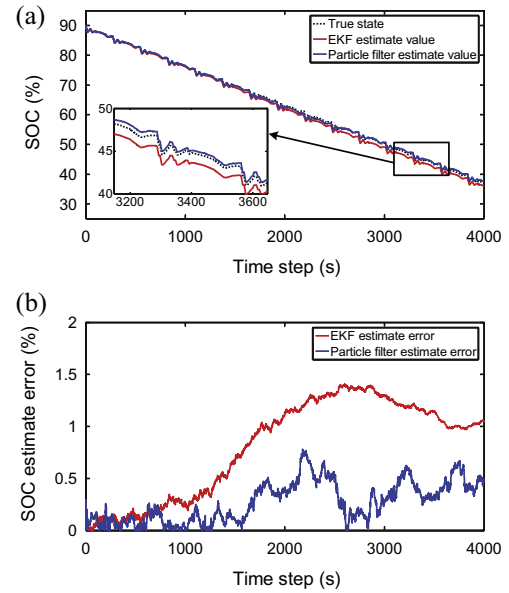
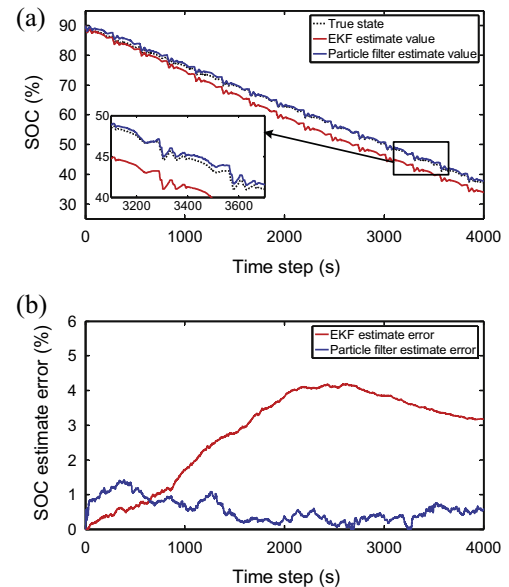
**Fig. 7.** Comparison of EKF and PF with a -5 mA drift current: (a) SOC estimation results of EKF and PF. (b) SOC estimate error of EKF and PF.**Fig. 8.** Comparison of EKF and PF with a -15 mA drift current: (a) SOC estimation results of EKF and PF. (b) SOC estimate error of EKF and PF.

Table 3Performance comparison of EKF and PF with a -5 mA and -15 mA drift current.

	RMSE ^a (%)	MAE ^b (%)	APET ^c (s)
<i>With a -5 mA drift current</i>			
EKF	1.849	1.405	0.315
PF	0.983	0.780	2.512
<i>With a -15 mA drift current</i>			
EKF	4.134	4.198	0.303
PF	0.926	1.421	2.621

^a Root-mean square error.^b Maximum absolute error.^c Average of program execution time.

As described in Eqs. (1) and (2), the cell SOC model can be depicted by Fig. 4(a). The parameters of the cell SOC model includes nominal capacity, initial SOC, coulombic efficiency, current and open-circuit voltage which is determined by cell voltage, internal resistance and current. Furthermore, we establish the battery pack model as shown in Fig. 4(b) which consists of single cell models and SOC, BMS, equalization topologies and algorithms.

3.3. The cell SOC-PF based equalization algorithm

Through the discretization of Eqs. (1) and (2), the discrete-time state space equations are described as Eq. (3):

$$\begin{cases} x_k = f(x_{k-1}, u_{k-1}) + w_{k-1} \\ y_k = g(x_k, u_k) + v_k \end{cases} \Leftrightarrow \begin{cases} SOC_k = SOC_{k-1} + (i_{k-1} + i_{E,k-1})\Delta t / C_N + w_{k-1} \\ E_k = E_0 - K_0 SOC_k - K_1 / SOC_k + K_2 \ln(SOC_k) + K_3 \ln(1 - SOC_k) \\ \quad - (i_{k-1} + i_{E,k-1})R + v_k \end{cases} \quad (3)$$

where x represents the state variable, y represents the observed variable, u represents the model input, Δt is the sample period, $\{w\}$ and $\{v\}$ are the Gaussian system and observation noises with covariance Qw and Qv , respectively.

Measurement noise is inevitable in SOC estimation. Since there are accumulated errors caused by current and voltage measurement noise, a SOC-particle filter (PF) based equalization algorithm is proposed. The PF methods have become a popular class of algorithms for solving the optimal estimation problems for non-linear

and non-Gaussian state space models [21]. The algorithm of the particle filter for SOC estimation is shown in Table 1.

4. Simulation and experiment verification

4.1. Accuracy of SOC-PF method

Considering the difficulty in massive calculation and data recording, analyzing the accuracy of all cell SOC is messy and without structure. Thus we chose one cell which has a close capacity to the average capacity of the pack for simulation experiments. The parameters of the battery model were determined by the recursive least-squares method based on the experimental data of the cell. The resulting values of the parameters are listed in Table 2. The estimated values from the battery model and the experimental values are compared in Fig. 5.

The initial SOC for simulation is set as 89%. The dynamic current condition is shown in Fig. 6 which is referred to the QCT 897-2011 [22]. All the simulations are performed on an Intel (R) Core (TM) i3-3220 CPU with a 4G RAM. To verify the robustness of the proposed method, a drift current is added in the simulation experiments. The comparisons of EKF and PF with a -5 mA and -15 mA drift currents are shown in Figs. 7 and 8, respectively. The root-mean square error (RMSE) and maximum absolute error (MAE) are calculated to compare the estimation performances of EKF and PF and the average of program execution time (APET) is counted to evaluate the on-line performance. The numerical results are shown in Table 3.

As shown in the figures and table, the PF has a relatively higher accuracy of SOC estimation than the EKF method. The RMSEs of EKF and PF with a -5 mA drift current are 1.849% and 0.983%, respectively. The MAEs are 1.405% and 0.780%, respectively. Both the RMSE and MAE of PF are lower than EKF, thus PF shows better estimation performance than the EKF method. Such was the case with the simulation experiment with a -15 mA drift current. The RMSE and MAE of EKF are 4.134% and 4.198%, respectively. The RMSE and MAE of PF are 0.926% and 1.421%, respectively. The PF has shown its strong robustness when the drift current increases.

As shown in Table 3, the APETs of EKF and PF are 0.315 s and 2.512 s with a -5 mA drift current. When the drift current is -15 mA, the APETs of EKF and PF are 0.303 s and 2.621 s, respectively. The particle number of PF is set as 250. The results show

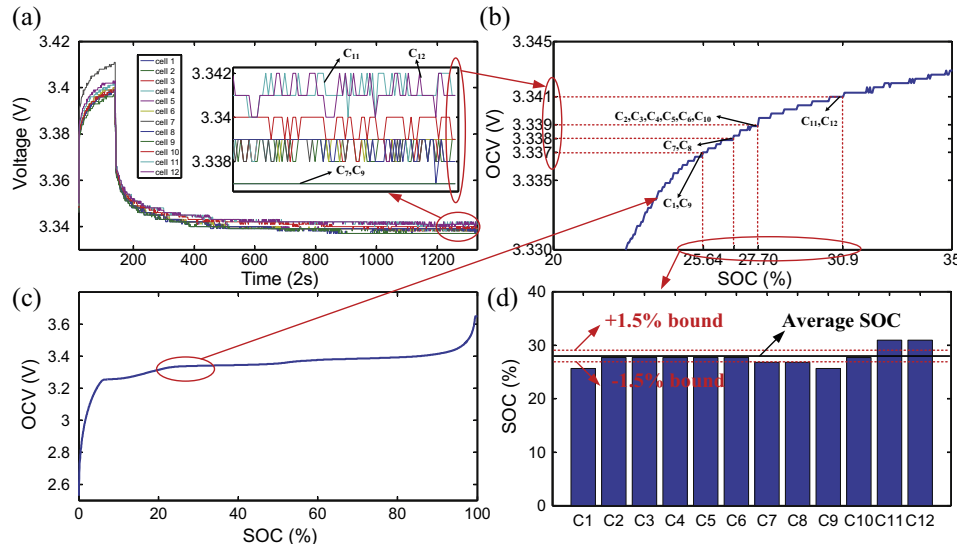


Fig. 9. Observation of cell SOC at the beginning of equalization: (a) cell voltage curves. (b) Look-up table of SOC-OCV. (c) SOC-OCV curve. (d) Cell SOC.

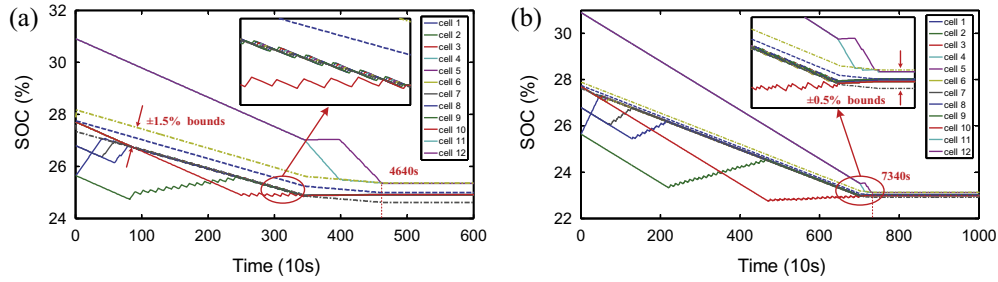


Fig. 10. Comparison of FLCC equalization algorithm: (a) FLCC with $\pm 0.5\%$ SOC bounds. (b) FLCC with $\pm 1.5\%$ SOC bounds.

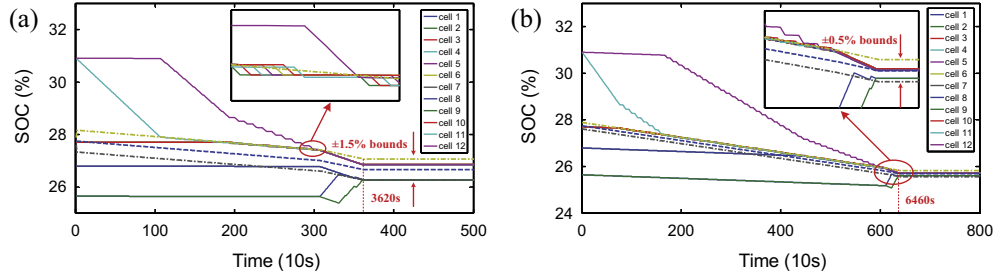


Fig. 11. Comparison of FHCD equalization algorithm: (a) FLCC with $\pm 0.5\%$ SOC bounds. (b) FLCC with $\pm 1.5\%$ SOC bounds.

Table 4

Performance comparison of FLCC and FHCD.

	Equalization time (s)	Standard deviation	Average SOC loss (%)
$\pm 1.5\%$ SOC bounds			
FLCC	4640	0.163	2.413
FHCD	3620	0.266	0.947
$\pm 0.5\%$ SOC bounds			
FLCC	7340	0.044	4.894
FHCD	6460	0.048	2.043

that the EKF method has a better real-time performance than the PF method. The particle number and computational complexity of PF needs to be reduced for on-line estimation.

4.2. Comparison of FLCC and FHCD equalization algorithms

Based on the equalization topology, the equalization algorithms can be divided into two types: the first low-SOC cell charge (FLCC) equalization algorithm and the first high-SOC cell discharge (FHCD) equalization algorithm. The FLCC equalization algorithm first lets the energy be transferred from the pack to the cells with lower SOC. After the weak cells receive the lower bound of the average SOC, the energy starts transforming from the most charged cells to the pack. On the contrary the FHCD equalization algorithm first allows the energy transfer from the most charged cells to the pack until the cells reduce to the higher bound of the average SOC.

Because of the flat OCV curves and voltage plateaus in large SOC ranges, the experiments are compatibly verified from 20% to 80% SOC range. The cells are first charged to 3.4 V, and then kept resting for 40 min as shown in Fig. 9(a). Fig. 9(c) shows the relationship between the SOC and the OCV which is measured by experiments. As shown in Fig. 9(b) the maximum difference of OCV is 5 mV and the cell SOC can be obtained by the SOC–OCV look-up table method. The initial SOC for equalization is shown in Fig. 9(d). The average cell SOC is 27.7383%.

The comparisons of FLCC and FHCD equalization algorithms are shown in Figs. 10 and 11. The numerical results are shown in Table 4. As can be seen from the equalization results, the equalization times of FLCC are 4640 s with $\pm 1.5\%$ SOC bounds and 7340 s

with $\pm 0.5\%$ SOC bounds. The time costs of FHCD are 3620 s with $\pm 1.5\%$ SOC bounds and 6460 s with $\pm 0.5\%$ SOC bounds. The equalization execution time of FHCD is shorter than the FLCC algorithm. The standard deviations with $\pm 1.5\%$ SOC bounds of FLCC and FHCD are 0.163 and 0.266, meanwhile the standard deviations with $\pm 0.5\%$ SOC bounds of FLCC and FHCD are 0.44 and 0.48, respectively. The FLCC equalization algorithm has better consistency than the FHCD equalization algorithm. From the perspective of SOC loss, the FLCC algorithm has a 2.413% average SOC loss with $\pm 1.5\%$ SOC bounds and a 4.894% average SOC loss with $\pm 0.5\%$ SOC bounds. Meanwhile the FHCD algorithm has a 0.947% average SOC loss with $\pm 1.5\%$ SOC bounds and a 2.043% average SOC loss with $\pm 0.5\%$ SOC bounds. The FHCD equalization algorithm is more suitable for energy saving systems.

5. Conclusion

This paper proposes a novel active equalization method based on the remaining capacity of cells which is suitable for lithium-ion batteries in EVs. The main concluding remarks can be made below:

- (1) The lithium-ion cell inconsistency has been analyzed based on the commercial cells from an urban electric bus. The CCVs based equalization method is easy to cause over-equalization due to the flat OCV curve and voltage plateaus in large SOC ranges. The SOC and capacity based equalization methods are more suitable for lithium-ion batteries.
- (2) A bidirectional transformer topology is introduced for active equalization. The battery pack model using for equalization of lithium-ion batteries is established based on cell models, cell SOC, BMS, equalization topologies and equalization algorithms.
- (3) The SOC-PF method is first employed to use in the capacity based equalization strategies. The drift current is introduced to verify the robustness of the proposed method. The SOC-PF has shown good accuracy in cell SOC estimation with RMSEs and MAEs less than 1% and 2%.

- (4) The FLCC and FHCD equalization algorithms are proposed and compared with $\pm 0.5\%$ and $\pm 1.5\%$ SOC bounds. The equalization results have shown that the FLCC equalization algorithm has better consistency than the FHCD equalization algorithm. The equalization execution time of FHCD is shorter than the FLCC algorithm and the FHCD equalization algorithm is more suitable for energy storage systems.

The SOC estimation method for cells and the equalization algorithms presented in this paper are reasonable and applicable for EVs.

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